Lecture 6.3: Machine Learning in Economic Research

AREC 570 / ECON 530 · Unit 6

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Before class

- Athey and Imbens, “The State of Applied Econometrics: Causality and Policy Evaluation” (Athey and Imbens 2017)
- Athey and Imbens, “Machine Learning Methods That Economists Should Know About” (Athey and Imbens 2019)
- Mullainathan and Spiess, “Machine Learning: An Applied Econometric Approach” (Mullainathan and Spiess 2017)

Readings and course handouts are available on Canvas.

Example papers in this lecture

- Background: Einav and Levin (2014); Bahoo et al. (2025)
- Overview for agribusiness: Maruejols et al. (2025)
- Prediction: Zhao et al. (2025)
- Prediction within a larger analysis: Li and Yu (2025); Belloni et al. (2014)
- Heterogeneous treatment effects: Stetter et al. (2022)
- Further reading: Bittmann's guide from OLS to causal machine learning (Bittmann 2025)

Machine learning in a nutshell

If you take away one idea today: A prediction method learns from observations we already have, then we check how well it predicts observations it has not seen

1. Choose the outcome to predict and the information available when a prediction is made
2. Fit a prediction rule using one group of observations
3. Compare its predictions with actual outcomes in a separate group
4. Report how much better it performs than a simple benchmark

In a causal study, machine learning can help account for differences between groups before estimating an effect

How does Facebook recognize faces in photos?

Algorithms use a large dataset of photos labeled as having a face or not to estimate a function $f(x)$ that predicts the presence y of a face from pixels x.

This similarity to econometrics raises questions:

- Are these algorithms merely applying standard techniques to novel and large datasets?
- If there are fundamentally new empirical tools, how do they fit with what we know?
- As empirical economists, how can we use them?

Source: Mullainathan and Spiess (2017), page 87.

Two goals: prediction and parameter estimation

Many economic applications estimate a parameter, such as a treatment effect or price sensitivity

Supervised machine learning methods are often designed to predict an outcome accurately for new observations

Economists can also use prediction methods within procedures that estimate effects

This lecture covers where these methods came from, the three roles they play in empirical economics, and how a methods section reports them

Source: Mullainathan and Spiess (2017), pages 87 to 88.

Evaluating predictions and parameters

	Prediction, $\widehat{y}$	**Parameter estimation, $\widehat{\beta}$**
Target	The outcome for a new observation	An effect or behavioral parameter
Common techniques	LASSO, random forests, and other prediction methods	OLS, IV, difference in differences, and structural models
Evaluation	Accuracy on data not used in estimation	Identification: the source of variation and its assumptions
Verifiability	Directly testable: predictions and outcomes are both observed	Rests on assumptions about the data-generating process

Source: Mullainathan and Spiess (2017), pages 88 and 92.

Background

Machine learning's use in applied economics grew over the past 20 years

Period	Development
1984 to 2001	Regression trees (Breiman et al., 1984), the LASSO (Tibshirani, 1996), and random forests (Breiman, 2001)
1990s	Neural networks studied in econometrics, but they "did not catch on at the time"
2001	Breiman's "two cultures": statistics relied on data models, while an algorithmic culture grew in other fields
2014	Early reviews for economists: Varian; Einav and Levin; Belloni, Chernozhukov, and Hansen
2017 to 2019	Mullainathan and Spiess; causal forests (Wager and Athey); Athey and Imbens's review for economists
2020 to 2025	Reviews and special issues in agricultural and applied economics: Storm et al. (2020); Maruejols et al. (2025)

Sources: Athey and Imbens (2019), pages 686 to 688, 696 to 698, and 701; Mullainathan and Spiess (2017), page 87; Storm et al. (2020); Maruejols et al. (2025).

Why economists adopted machine learning

1. **More readily available access to “big data”**
    - Properties: many units, many variables per unit, and unstructured sources such as satellite images and text
    - Challenge organizing and studying this type of data (or even to decide which variables to construct) without a simplifying conceptual framework
    - Better data allow for (1) sharper tests of existing models, (2) tests of theories that had previously been difficult to assess, and (3) assistance choosing controls or functional forms
2. **Prediction questions**
    - Some policy problems require an accurate forecast and no causal parameter

Sources: Athey and Imbens (2019), pages 686 to 687; Mullainathan and Spiess (2017), pages 88, 98, and 102, Einav and Levin (2014)

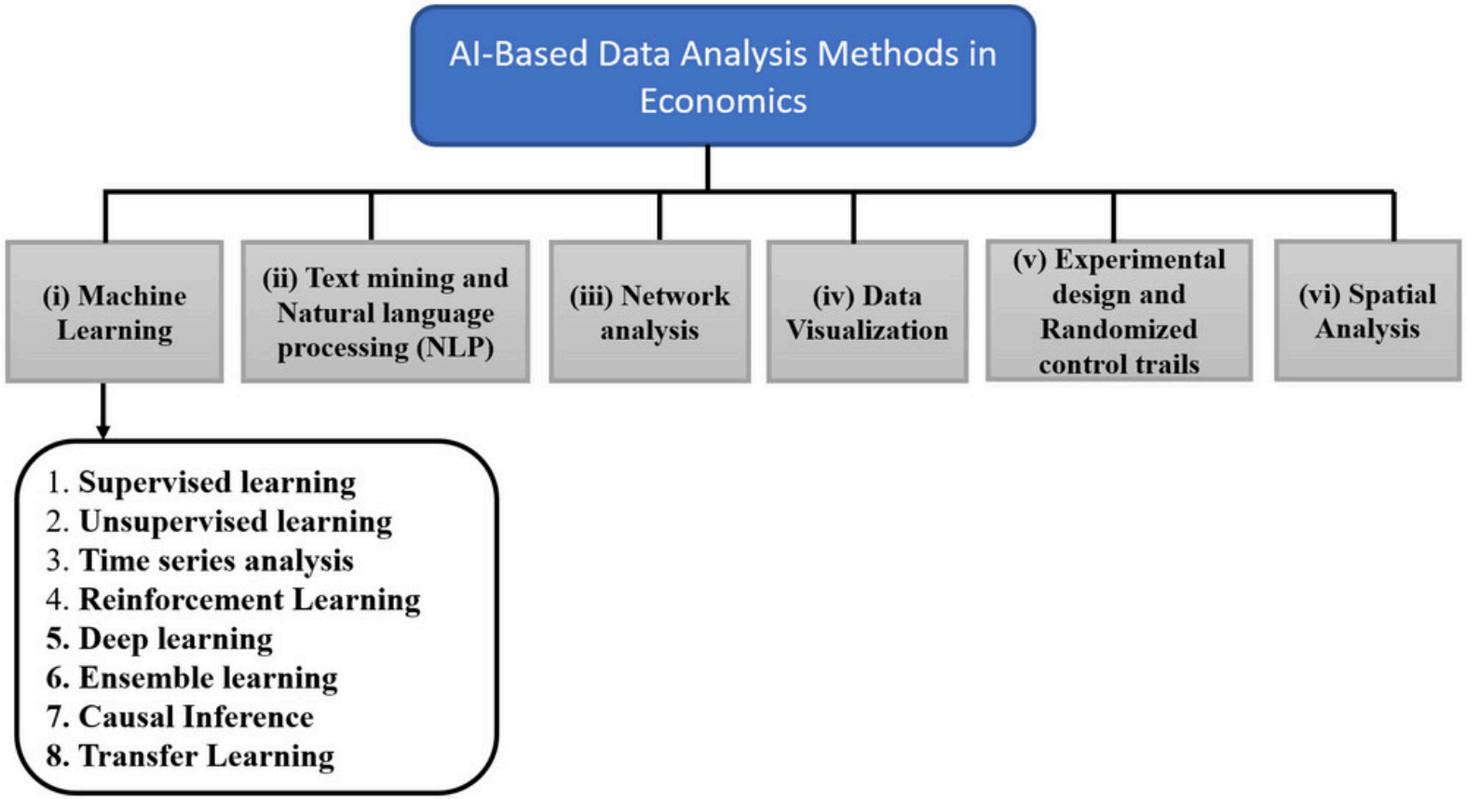
Why adoption was slow

Economics journals emphasize formal properties of estimators, such as consistency and asymptotic normality, which early machine learning methods did not provide

However, recent work established inference for adapted methods, including double selection and causal forests

Sources: Athey and Imbens (2019), pages 686 to 687; Mullainathan and Spiess (2017), pages 88, 98, and 102.

Machine learning within the AI-based tool landscape



Source: Bahoo et al. (2025), Section 4.3 and Figure 6.

How machine learning works

How a house-price prediction rule learns

Imagine we have recent home sales and want to predict the sale price of another house

1. For each sold house, record its sale price and characteristics such as size and location
2. Set aside some houses before fitting a model
3. Use the remaining houses to fit a rule that turns characteristics into predicted prices
4. Measure each mistake by comparing a prediction with the actual sale price

For a house that sold for **$300,000**, a prediction of **$320,000** misses by **$20,000**

Is this rule sufficient, or should it be adjusted to reduce the average mistake on the houses used for fitting?

Source: Mullainathan and Spiess (2017), page 89

How do we score a house-price prediction?

Suppose a house sold for **$300,000** and a rule predicted **$320,000**

Loss function	Score for this house	What it emphasizes
Absolute error	$\|320 - 300\| = 20$ thousand dollars	The size of each miss
Squared error	$(320 - 300)^2 = 400$ squared thousand dollars	Large misses receive more weight

A **loss function** assigns a score to each prediction error. We choose a rule to reduce average loss across the training houses

Source: Mullainathan and Spiess (2017), page 89

Why we keep some houses aside

Expected loss is average loss on future houses. We cannot calculate it directly because their sale prices are unknown

1. **Train:** choose the rule that reduces average loss on the houses used for fitting, called empirical loss
2. **Test:** predict the prices of the houses set aside earlier
3. **Compare:** use their average loss to estimate performance on new houses

The next paper reports held-out R^2 as a summary of predictive fit. Higher held-out R^2 means better predictions relative to predicting the same average for every house

Source: Mullainathan and Spiess (2017), pages 89 to 90

An example: predicting house values with OLS and machine learning

Table 1

Performance of Different Algorithms in Predicting House Values

Method	*Prediction performance (R^2)*		*Relative improvement over ordinary least squares by quintile of house value*				
	Training sample	*Hold-out sample*	1st	2nd	3rd	4th	5th
Ordinary least squares	47.3%	41.7% [39.7%, 43.7%]	–	–	–	–	–
Regression tree tuned by depth	39.6%	34.5% [32.6%, 36.5%]	−11.5%	10.8%	6.4%	−14.6%	−31.8%
LASSO	46.0%	43.3% [41.5%, 45.2%]	1.3%	11.9%	13.1%	10.1%	−1.9%
Random forest	85.1%	45.5% [43.6%, 47.5%]	3.5%	23.6%	27.0%	17.8%	−0.5%
Ensemble	80.4%	45.9% [44.0%, 47.9%]	4.5%	16.0%	17.9%	14.2%	7.6%

1. Start with 51,808 homes from the 2011 American Housing Survey, including each home's value and 150 characteristics
2. Use 10,000 homes to fit OLS and several machine learning methods
3. Ask each fitted method to predict values for the other 41,808 homes
4. Compare those predictions with the homes' observed values using R^2

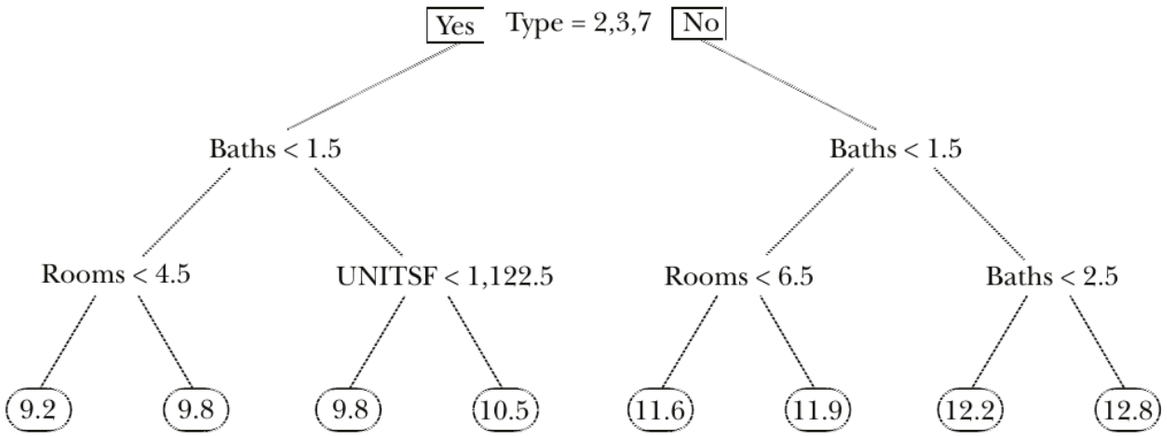
The random forest explains **45.5%** of the variation in held-out values, compared with **41.7%** for OLS. Its training R^2 was **85.1%**

Source: Mullainathan and Spiess (2017), pages 89 to 90 and Table 1.

How a regression tree predicts

Figure 1

A Shallow Regression Tree Predicting House Values



Note: Based on a sample from the 2011 American Housing Survey metropolitan survey. House-value predictions are in log dollars.

1. Start with all houses in one group
2. Find the single yes-or-no split, on one variable, that best separates high-value from low-value houses (here, house type)
3. Repeat the search within each new group (bathrooms, then rooms or square footage)
4. Stop at a set depth; the prediction for a house in a final group (a leaf) is that group's average value

Source: Mullainathan and Spiess (2017), pages 90 to 91 and Figure 1.

Why stop a tree from dividing further?

1. A tree can keep dividing the training houses into smaller groups
2. With enough divisions, it can nearly memorize their sale prices
3. Those very specific groups may predict poorly for houses the tree has not seen
4. We limit the number of divisions and require several houses in each final group

Fitting the training houses closely while predicting new houses poorly is called **overfitting**

Source: Mullainathan and Spiess (2017), pages 91 to 93 and Table 2.

How do we choose when a tree should stop?

1. Keep the test houses aside until the final evaluation
2. Divide the training houses into several groups
3. Fit a short tree on all but one group, then check its predictions for the group left out
4. Repeat with each group left out, then repeat the process for deeper trees
5. Choose the depth with the smallest average prediction error
6. Fit a tree of that depth on all training houses and evaluate it once on the test houses

This repeated check within the training data is called **cross-validation**

Sources: Mullainathan and Spiess (2017), pages 92 to 93; Bittmann (2025), Section 3.

How do the methods differ?

Each method fits a rule on training data and checks it on held-out data

Method	Prediction rule	How flexibility is limited
OLS	Weighted sum of characteristics	Chosen characteristics and interactions
Regression tree	Yes-or-no splits	Maximum splits and minimum group size
LASSO	Weighted sum of characteristics	Some weights shrink to zero
Random forest	Average of many trees	Limits on each tree and averaging

Sources: Mullainathan and Spiess (2017), pages 92 to 94 and Table 2; Athey and Imbens (2019), pages 690 and 695 to 699.

Examples of how machine learning is used in empirical economics

1. Prediction as the objective: Predicts an outcome, such as household food insecurity

2. Prediction within a larger analysis: Selects the variables that enter an economic model or a treatment-effect regression

3. Heterogeneous treatment effects: Estimates how a treatment effect varies across farms or households

Sources: Mullainathan and Spiess (2017); Athey and Imbens (2019), pages 686 to 688; Maruejols et al. (2025).

Role 1: Prediction as the objective

Zhao et al. (2025): predicting which households are calorie poor

The question is whether information a program could collect can identify rural households with inadequate calorie intake

1. Use three days of food records to calculate calorie intake for each of 1,080 households
2. Mark a household as calorie poor if its intake falls below the study's threshold
3. Record household, village, and food-market characteristics that might help predict that outcome
4. Fit a random forest, LASSO, and a logit model using 756 households
5. Predict calorie poverty for the other 324 households and compare predictions with their measured outcomes

Source: Zhao et al. (2025), Sections 3 to 5.

How much does the random forest improve prediction?

Model (test sample)	Accuracy
Random forest	65.7%
LASSO	61.1%
Logit benchmark	58.9%

Accuracy is the share of test households classified correctly. The forest exceeds the logit benchmark by 6.8 percentage points

Source: Zhao et al. (2025), Sections 3 to 5 and Table 3

What can a predictor ranking tell us?

Zhao et al. find that household income and access to food markets help predict calorie poverty

- A predictor can be useful because it moves with another factor that the data do not measure
- Correlated predictors can replace one another, so the variables chosen by LASSO may change when the training sample changes
- The ranking tells us which information helps classify households in this dataset

Sources: Zhao et al. (2025), Section 5.2; Mullainathan and Spiess (2017), page 97, Figure 2; Stetter et al. (2022), page 740, footnote 17.

Role 2: Prediction within a larger analysis

Li and Yu (2025): selecting apple attributes

The question is how much consumers value apple attributes such as origin, size, and production method

1. Ask 100 consumers to choose among two apples and a no-purchase option in four scenarios
2. Record each apple's attributes and price in every choice
3. Use LASSO to select attributes that help predict choices, while always retaining price
4. Fit a choice model with the retained attributes and calculate willingness to pay from its coefficients

Source: Li and Yu (2025), Sections 2 and 3.

Why are Organic and Japan omitted?

LASSO shrinks attribute coefficients toward zero. The authors choose the penalty using **tenfold cross-validation**

At the chosen penalty, **Organic** and **Japan** reach zero. They contribute the weakest additional signals for predicting choices after the other recorded attributes are considered

Price stays in the model because the authors exempt it from the penalty. The price coefficient is needed to calculate willingness to pay

This selection describes the fitted model for these respondents. It does not establish that every respondent ignored either attribute

Source: Li and Yu (2025), Section 3.1, Figure 2, and Table 4

LASSO changes willingness-to-pay estimates

Willingness to pay relative to China, in euros per kg

Apple attribute	All attributes	After LASSO
France	1.20	1.53
Italy	1.42	1.40
Small apples	1.30	1.41
Organic	−0.04	excluded

- **All attributes:** a conditional logit with every attribute
- **After LASSO:** the same model without the two attributes the LASSO excluded (Organic and Japan)

Source: Li and Yu (2025), Section 3, Tables 6 and 7.

Why select controls for both equations?

Belloni et al. revisit the relationship between **earlier abortion rates** and **later crime rates** across states

A state characteristic may be related to both rates. For example, an initial state condition could predict how abortion rates change and how crime rates change over time

If controls are selected only for predicting crime, the procedure might omit a variable that strongly predicts abortion rates and also matters for crime

Double selection retains controls selected from either prediction step before estimating the abortion-rate coefficient

Source: Belloni et al. (2014), Sections 1 and 7

Belloni et al. (2014): double selection in practice

The application has **576 state-year observations** and **251 candidate controls**, including initial conditions, interactions, and trends

1. Use LASSO to find controls that predict the relevant abortion rate
2. Use LASSO again to find controls that predict the crime rate
3. Combine the controls selected in either step
4. Regress the crime rate on the abortion rate and those controls using OLS

The final models include **8 to 14** of the 251 candidate controls, depending on the crime outcome

The estimates for violent crime and murder become imprecise after the selected controls are included

Source: Belloni et al. (2014), Sections 1, 2.2, and 7.

Role 3: Heterogeneous treatment effects

How does prediction help estimate an effect?

Suppose larger farms are more likely to join a program and also use more fertiliser. A direct comparison of participants and other farms mixes those farm differences with the program effect

1. Use farm characteristics to predict participation and expected fertiliser use
2. Account for those expected differences when comparing participants with other farms
3. Use many trees to find groups of farms where the adjusted comparison differs

The result estimates an **average program effect for farms with similar characteristics**

Sources: Athey and Imbens (2019), pages 701 to 704; Stetter et al. (2022), Sections 2.3 and 4.1 to 4.3.

Stetter et al. (2022): do program effects differ across farms?

The study asks whether participation in an agri-environment scheme changes farm outcomes, and whether the change differs by farm characteristics

1. Start with 2,758 Bavarian farms in 2014, including 1,641 program participants
2. Record farm and county characteristics, participation, and four environmental outcomes
3. Use the characteristics to predict which farms participate and what outcomes to expect
4. Account for those predicted differences when comparing participants with other farms
5. Use a causal forest to estimate how the effect varies across groups of farms

Source: Stetter et al. (2022), Sections 2.3, 4.1, and 4.2.

Which farms have estimated benefits from the schemes?

Outcome	Share with a detectable improvement
Fertiliser expenditure	33%, a reduction of €14 per hectare among them
Land-use diversity	55%, a small increase
Pesticide expenditure	7%
Emissions	Under 1%; about 4% show detectable increases

Estimated effects vary by location, farm size, and yield potential. The authors use these differences to discuss how payments might be targeted

Source: Stetter et al. (2022), Sections 5.2, 5.3, and 6 and Table 3.

How do the authors support the comparison?

The causal forest compares farms after accounting for their recorded characteristics. The authors examine two conditions

1. **Comparable farms:** after accounting for recorded characteristics, participation is as good as randomly assigned
2. **Both options are possible:** farms with similar characteristics could participate or stay out of the program

The authors exclude 23 farms that lack a suitable comparison and check how their findings respond to alternative assumptions about missing factors

Source: Stetter et al. (2022), Sections 4.3 and 5.1.

Reporting a machine learning method

What to report in a machine learning methods section

Item	Example
1. What the method does in the study	Zhao et al. use it to predict calorie poverty
2. What is predicted and which information is used	Calorie poverty from household and market characteristics
3. Which observations are used for fitting and testing	70% training, 30% test
4. How method settings are chosen	Cross-validation; number of trees
5. How well it performs on test observations	Classification accuracy, compared with logit
6. If effects are claimed, what supports the causal comparison	Stetter et al. discuss comparable farms and overlap
7. How another researcher could repeat the analysis	Software, package versions, random seed, and code

Sources: Zhao et al. (2025); Li and Yu (2025); Stetter et al. (2022); Athey and Imbens (2019), pages 686 to 687.

What to consider for your proposal

- **The role:** does the question concern predicting an outcome, selecting variables, or how an effect varies?
- **The claim:** does the conclusion require a causal effect? If so, what is the design, and where does machine learning enter it?
- **The benchmark:** which simpler model or rule should the method outperform?
- **The evidence:** which out-of-sample measure, or which identification check, will the paper report?

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Course navigation

- All lectures
- Schedule and preparation
- Course reading list
- Empirical strategy checklist
- Methods map
- Bittmann (2025), a practical guide from OLS to causal machine learning
- Download this lecture as a PDF